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# 1. Introduction

This document models connected coffee machines that send telemetry to a `central cloud site`, receive OTA firmware updates, and produce operational and audit logging evidence. Authored functional architecture remains stable while runtime, code, and verification layers are inferred from infrastructure, source, and test artifacts.

## 1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

# 2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

## 3. Document Health Snapshot

| View | Authored Status | Authored Status Explanation | Units in Scope | Direct Evidence Coverage | Gap Count | Heuristic Confidence | Heuristic Basis |
|---|---|---|---|---|---|---|---|
| Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`) | stable | Stable authored capability boundaries and responsibility ownership. | 7 | 7 | 0 | high | Defaults authored units to covered; use authoredStatus as the primary publication signal. |
| Communication View (`VIEW-COMMUNICATION`) | in-review | Communication handoffs are mapped; protocol detail remains partially inferred. | 7 | 5 | 2 | medium | Counts a unit as covered when direct inferred runtime evidence exists. |
| Deployment View (`VIEW-DEPLOYMENT`) | in-review | Deployment/runtime mapping is present with evolving control-plane evidence depth. | 7 | 5 | 2 | medium | Counts a unit as covered when direct inferred runtime evidence exists. |
| Security View (`VIEW-SECURITY`) | draft | Core threats are modeled while additional control evidence is being expanded. | 7 | 7 | 1 | high | Counts a unit as covered when explicit authored attack-vector mapping exists. |
| Traceability View (`VIEW-TRACEABILITY`) | in-review | Requirement mappings exist; verification and evidence coverage continue to be hardened. | 7 | 7 | 2 | high | Counts a unit as covered when direct inferred runtime or code evidence exists. |

## 4. Terms and Definitions

| Term | Definition |
|---|---|
| `actor` (EM-ACTOR) | A person or operational role that interacts with functional units. |
| `attack vector` (EM-ATTACK-VECTOR) | A technical misuse or attack path that targets functional, referenced, or runtime elements. |
| `audit log record` (DATA-AUDIT-LOG-RECORD) | Immutable log record for security and compliance analysis. |

| Term | Definition |
| --- | --- |
| `audit logging is enabled` (FEAT-AUDIT-LOGGING) | Feature flag enabling audit-grade log persistence. Aliases:audit enabled |
| `authored mapping` (EM-AUTHORED-MAPPING) | An explicit relationship declared in architecture inputs between authored or referenced elements. |
| `barista` (ACT-BARISTA) | Operates coffee machines and observes local machine health. Aliases:store operator |
| `brew cycle completed event is received` (EVT-BREW-CYCLE-COMPLETED) | Event raised when a coffee machine completes a brew cycle. Aliases:brew completed |
| `brew telemetry` (DATA-BREW-TELEMETRY) | Machine telemetry including brew duration, temperature, and status metadata. |
| `cloud control` (FG-CLOUD-CONTROL) | Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Aliases:central cloud site |
| `cloud ingest endpoint is unavailable` (EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE) | Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion. Aliases:ingest endpoint unavailable |
| `cloud logging service` (REF-CLOUD-LOGGING-SERVICE) | Managed log and metrics service used for fleet observability. Aliases:logging service |
| `cloud runtime operations` (FU-CLOUD-RUNTIME-OPERATIONS) | Operates cloud runtime deployment and scaling controls. Aliases:runtime operations |
| `cloud runtime sdk` (REF-CLOUD-RUNTIME-SDK) | SDK used by cloud units for runtime integrations. Aliases:cloud sdk |
| `code element` (EM-CODE-ELEMENT) | An inferred code structure or ownership element discovered from source trees and build metadata. |
| `coffee fleet system` (SYS-COFFEE-FLEET-SYSTEM) | Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture. Aliases:the coffee fleet system |
| `coffee machine fleet operation is active` (STATE-FLEET-OPERATION-ACTIVE) | State where machines and central cloud are actively serving operations. Aliases:fleet operation active |
| `device identity and secrets` (FU-DEVICE-IDENTITY-SECRETS) | Manages device identity material and secure configuration. Aliases:device secrets |
| `firmware bundle store` (REF-FIRMWARE-BUNDLE-STORE) | Object store containing signed OTA firmware bundles. Aliases:firmware store |

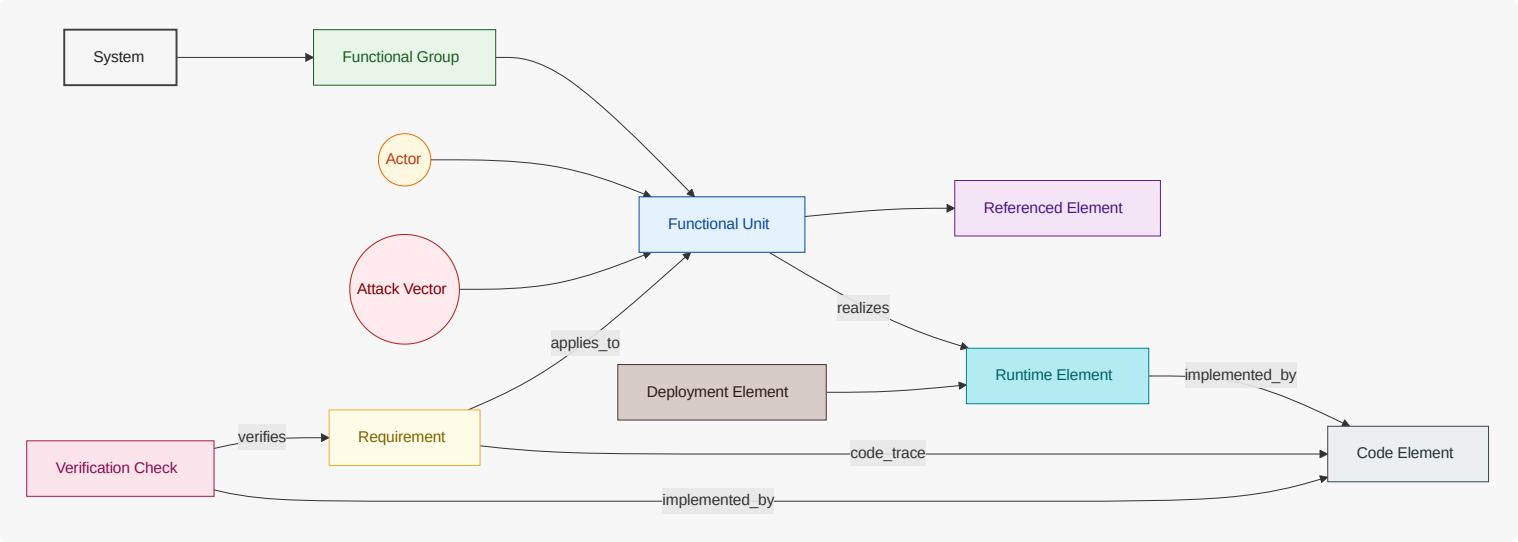
| Term | Definition |
| --- | --- |
| `firmware signature is valid` (COND-FIRMWARE-SIGNATURE-VALID) | Condition indicating firmware signature verification succeeded. Aliases:signature valid |
| `firmware signer service` (REF-FIRMWARE-SIGNER-SERVICE) | Service that signs firmware artifacts before OTA rollout. Aliases:signer service |
| `firmware validation failed event is received` (EVT-FIRMWARE-VALIDATION-FAILED) | Event raised when signature/hash validation fails on firmware. Aliases:validation failed |
| `fleet ingestion api` (FU-FLEET-INGESTION-API) | Ingests telemetry and machine status from edge devices. Aliases:ingestion api |
| `fleet observability reporting` (FU-FLEET-OBSERVABILITY-REPORTING) | Aggregates telemetry and update outcomes into operational and audit signals. Aliases:observability reporting |
| `fleet operator` (ACT-FLEET-OPERATOR) | Owns OTA rollout campaigns and fleet-level reliability. Aliases:fleet admin |
| `functional group` (EM-FUNCTIONAL-GROUP) | A major authored capability area that groups related functional units. |
| `functional unit` (EM-FUNCTIONAL-UNIT) | An authored working unit inside a functional group that owns specific behavior. |
| `inferred mapping` (EM-INFERRED-MAPPING) | A discovered relationship that links inferred runtime/code elements upward to authored design. |
| `iot ingest endpoint` (REF-IOT-INGEST-ENDPOINT) | Cloud endpoint for receiving telemetry and status from devices. Aliases:iot endpoint |
| `log tampering` (AV-LOG-TAMPERING) | Attempts to suppress or alter security/operational log records. Aliases:audit tampering |
| `machine edge` (FG-MACHINE-EDGE) | Capability area for device-local telemetry and OTA execution behavior. Aliases:coffee machine edge |
| `machine telemetry collection` (FU-MACHINE-TELEMETRY-COLLECTION) | Collects brew/device telemetry and publishes cloud-bound payloads. Aliases:telemetry collection |
| `malicious firmware bundle` (AV-MALICIOUS-FIRMWARE-BUNDLE) | Tampered firmware payload attempts to execute on devices. Aliases:firmware tampering |
| `mqtt device sdk` (REF-MQTT-DEVICE-SDK) | SDK used by edge components for telemetry and command transport. Aliases:mqtt sdk |

| Term | Definition |
| --- | --- |
| `normal mode is active` (MODE-NORMAL) | Default operation mode for telemetry and OTA behavior. Aliases:normal mode |
| `ota campaign is enabled` (FEAT-OTA-CAMPAIGN) | Feature flag enabling centralized OTA rollout behavior. Aliases:ota enabled |
| `ota campaign policy` (DATA-OTA-CAMPAIGN-POLICY) | Policy describing rollout cohorts, windows, and rollback criteria. |
| `ota campaign scheduled event is received` (EVT-OTA-CAMPAIGN-SCHEDULED) | Event raised when a fleet OTA campaign is scheduled. Aliases:campaign scheduled |
| `ota status` (DATA-OTA-STATUS) | Device-level status for firmware download, apply, and rollback outcomes. |
| `ota status reported event is received` (EVT-OTA-STATUS-REPORTED) | Event raised when device reports OTA apply status. Aliases:ota status reported |
| `ota update agent` (FU-OTA-UPDATE-AGENT) | Downloads, validates, and applies signed firmware updates on machines. Aliases:device update agent |
| `platform operations` (FG-PLATFORM-OPERATIONS) | Capability area for runtime operations and device identity/secrets lifecycle. Aliases:fleet platform operations |
| `platform operator` (ACT-PLATFORM-OPERATOR) | Owns runtime deployment and operational controls. Aliases:platform ops |
| `referenced element` (EM-REFERENCED-ELEMENT) | An architecture-relevant external, platform, or third-party dependency represented by role. |
| `requirement` (EM-REQUIREMENT) | A structured requirement statement that defines expected system behavior and maps to functional ownership. |
| `runtime element` (EM-RUNTIME-ELEMENT) | An inferred runtime realization element discovered from infrastructure and deployment sources. |
| `security analyst` (ACT-SECURITY-ANALYST) | Reviews identity, firmware integrity, and logging controls. Aliases:security reviewer |
| `spoofed device identity` (AV-SPOOFED-DEVICE-IDENTITY) | Unauthorized actor impersonates a machine identity. Aliases:device spoofing |
| `telemetry ingested event is received` (EVT-TELEMETRY-INGESTED) | Event raised when cloud ingestion accepts telemetry. Aliases:telemetry ingested |

| Term | Definition |
|---|---|
| `telemetry replay abuse` (AV-TELEMETRY-REPLAY-ABUSE) | Replayed telemetry floods ingestion and distorts reporting. Aliases:telemetry replay |
| `telemetry replay abuse is detected` (EVT-TELEMETRY-REPLAY-ABUSE-DETECTED) | Event raised when replayed telemetry behavior is detected and flagged as abuse. Aliases:telemetry replay detected |
| `update campaign orchestration` (FU-UPDATE-CAMPAIGN-ORCHESTRATION) | Selects target machine groups and coordinates OTA rollout phases. Aliases:campaign orchestration |
| `upward linking` (EM-UPWARD-LINKING) | Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. |
| `verification check` (EM-VERIFICATION-CHECK) | An inferred or authored verification artifact that validates requirement behavior with test evidence. |

## 4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.




# 5. Architecture Intent View

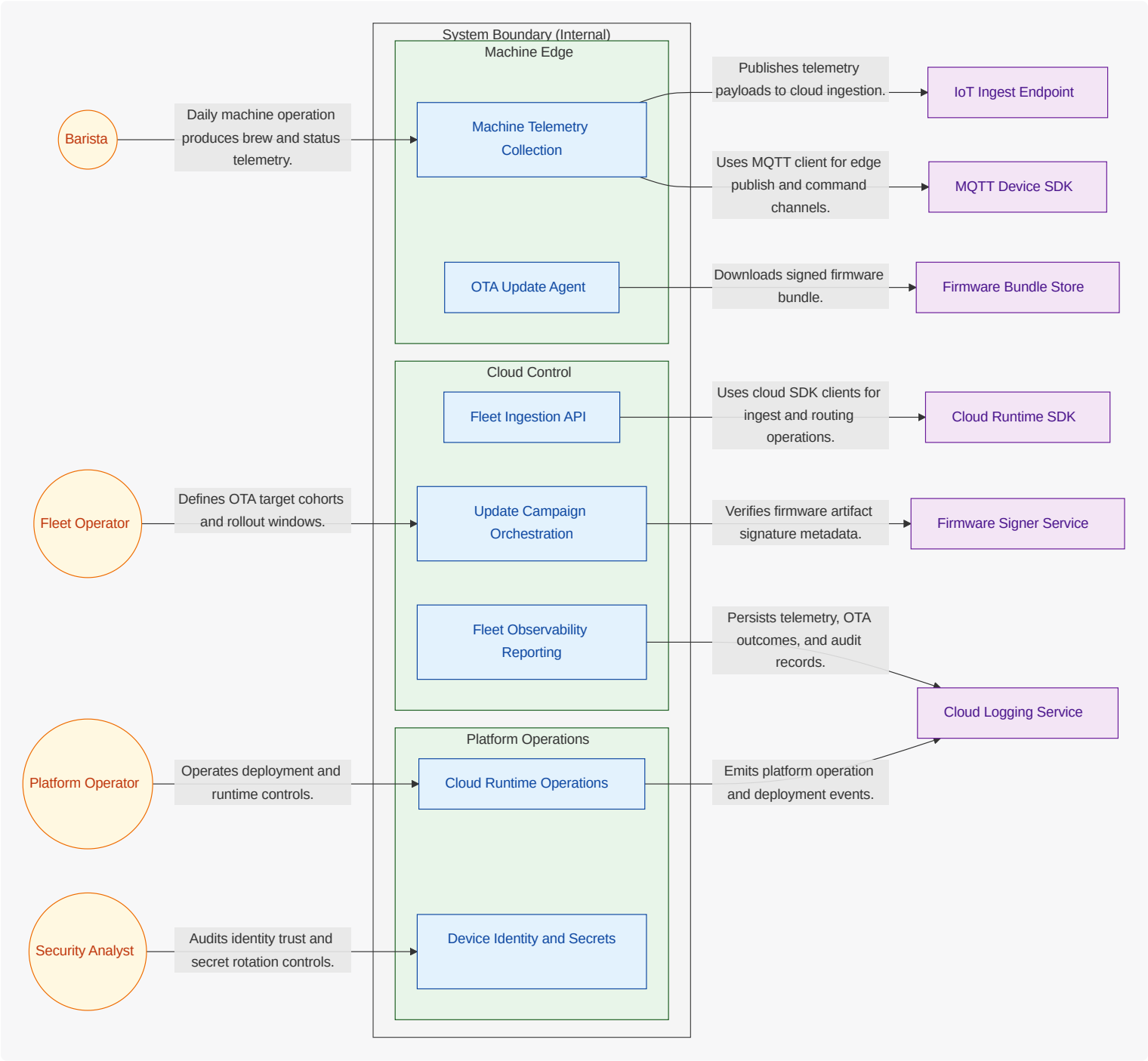
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

## 5.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

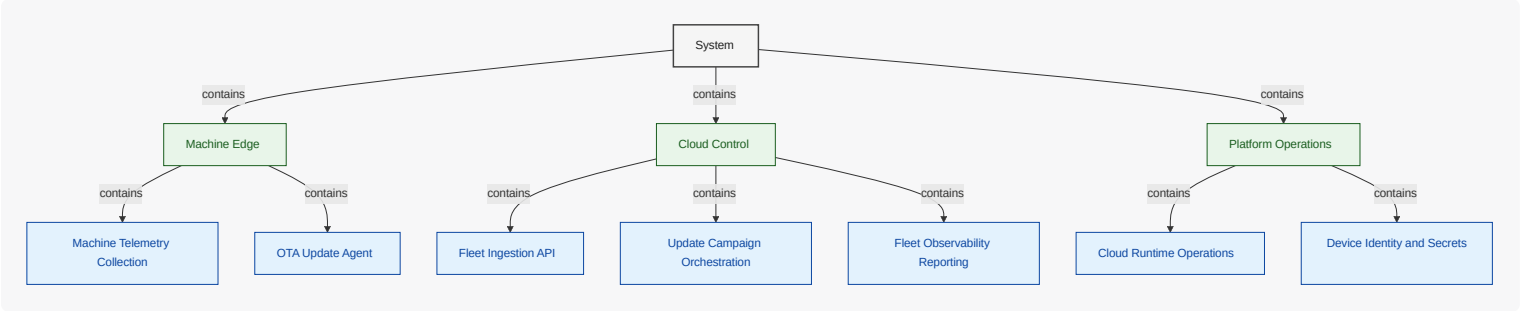
## 5.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.




## 5.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.


System
contains
contains
contains
Machine Edge
Cloud Control
Platform Operations
contains
contains
contains
contains
contains
contains
contains
Machine Telemetry Collection
OTA Update Agent
Fleet Ingestion API
Update Campaign Orchestration
Fleet Observability Reporting
Cloud Runtime Operations
Device Identity and Secrets

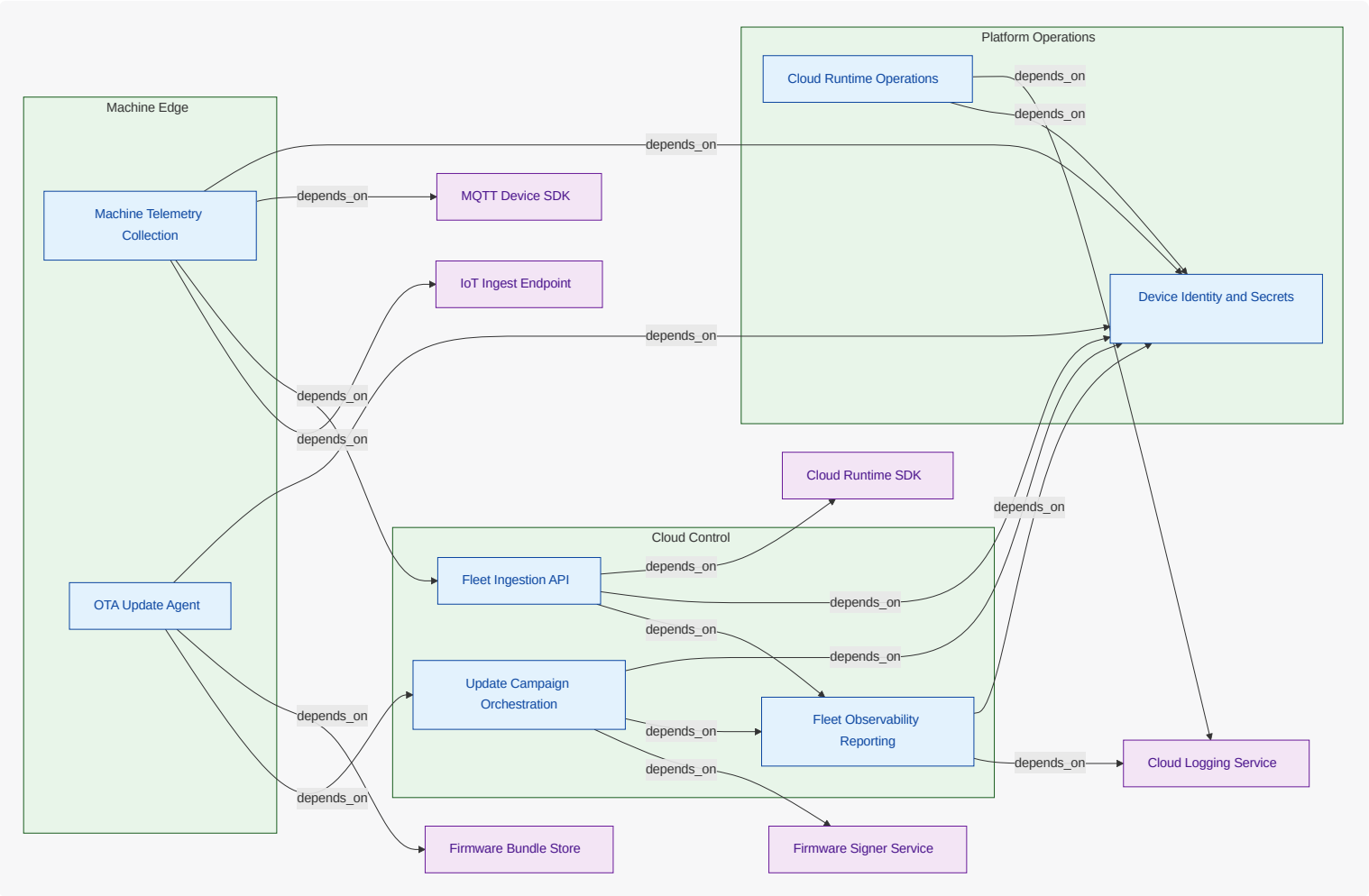
## 5.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

| | | |
|---|---|---|
| | Fleet Ingestion API | |
| Machine Telemetry Collection | Fleet Observability Reporting | Cloud Runtime Operations |
| OTA Update Agent | Update Campaign Orchestration | Device Identity and Secrets |
| Machine Edge | Cloud Control | Platform Operations |

## 5.5. Functional Unit Dependency Diagram

What this diagram shows: dependency paths between functional units (grouped in Functional Group containers) and key referenced elements. Arrow meaning: `A -→|depends_on| B` means unit `A` requires behavior or data from `B` to fulfill its responsibility.




## 5.6. Functional Groups and Units (View Scoped)

### 5.6.1. Machine Edge (FG-MACHINE-EDGE)

Device-local telemetry and OTA execution responsibilities.

`Machine Edge` owns behavior that executes directly on coffee machines. It gathers telemetry, applies OTA updates, and reports local status to cloud boundaries.

### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Collects brew/device telemetry and transmits cloud-bound telemetry payloads.

#### Owned Responsibility

functional responsibility for `machine telemetry collection`; includes decision and orchestration flow to `Device Identity and Secrets` and decision and orchestration flow to `Fleet Ingestion API`

#### Key Collaborators

`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

#### Linked Requirements

`REQ-COF-001`, `REQ-COF-006`

### OTA Update Agent (FU-OTA-UPDATE-AGENT)

Downloads signed firmware bundles, validates integrity, applies updates, and reports status.

#### Owned Responsibility

functional responsibility for `ota update agent`; includes decision and orchestration flow to `Device Identity and Secrets` and decision and orchestration flow to `Firmware Bundle Store`

#### Key Collaborators

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

#### Linked Requirements

`REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`

## 5.6.2. Cloud Control (FG-CLOUD-CONTROL)

Central cloud ingestion, campaign control, and fleet reporting.

`Cloud Control` owns central APIs and coordination logic for fleet telemetry and OTA rollouts. It provides global fleet visibility and deterministic rollout decisions.

### Fleet Ingestion API (FU-FLEET-INGESTION-API)

Receives telemetry and machine status, applies validation, and emits ingestion events.

#### Owned Responsibility

functional responsibility for `fleet ingestion api`; includes decision and orchestration flow to `Cloud Runtime SDK` and decision and orchestration flow to `Device Identity and Secrets`

#### Key Collaborators

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

#### Linked Requirements

`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`

### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Aggregates telemetry and OTA outcomes into dashboards, alerts, and audit records.

#### Owned Responsibility

functional responsibility for `fleet observability reporting`; includes decision and orchestration flow to `Cloud Logging Service` and decision and orchestration flow to `Device Identity and Secrets`

#### Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

#### Linked Requirements

`REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008`

### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Plans OTA rollout cohorts, drives update commands, and handles rollback flow decisions.

#### Owned Responsibility

functional responsibility for `update campaign orchestration`; includes decision and orchestration flow to `Device Identity and Secrets` and decision and orchestration flow to `Firmware Signer Service`

#### Key Collaborators

`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

#### Linked Requirements

`REQ-COF-003`, `REQ-COF-005`

### 5.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Cloud `runtime operations` plus identity and secret-management controls.

`Platform Operations` owns deployability, runtime guardrails, and secure identity/config lifecycle. It ensures cloud and device trust boundaries remain enforceable over time.

### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

#### Owned Responsibility

functional responsibility for `cloud runtime operations`; includes decision and orchestration flow to `Cloud Logging Service` and decision and orchestration flow to `Device Identity and Secrets`

#### Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

#### Linked Requirements

`REQ-COF-008`

### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

#### Owned Responsibility

functional responsibility for `device identity and secrets`

### Key Collaborators

none

### Linked Requirements

`REQ-COF-004`, `REQ-COF-007`

# 6. Communication View

Show who communicates with whom across key request paths, including interface edges and dependency handoffs.

## 6.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request flow?

## 6.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no direct inferred runtime/deployment evidence yet.
- `Device Identity and Secrets` has no direct inferred runtime/deployment evidence yet.

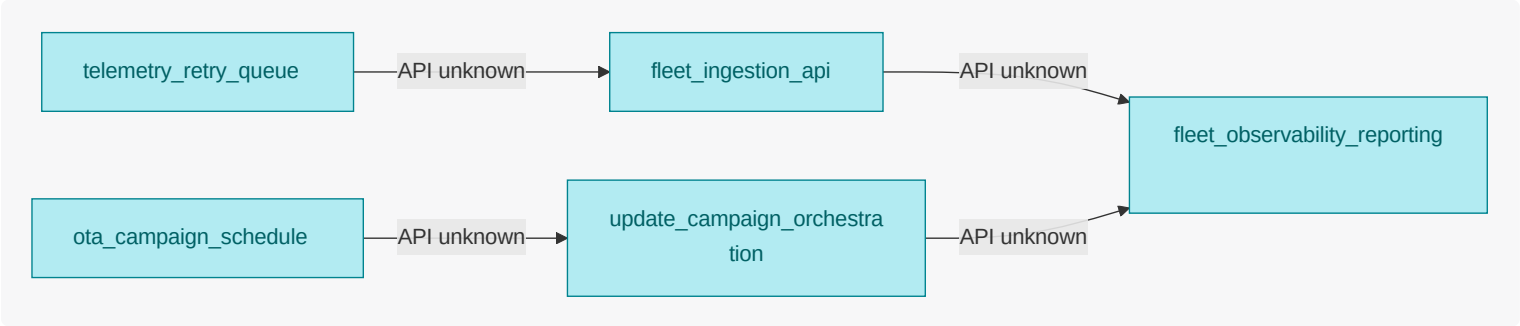
## 6.3. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/event/callback) to `authored mappings` where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

## 6.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE** This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.




## 6.5. Functional Groups and Units (View Scoped)

### 6.5.1. Machine Edge (FG-MACHINE-EDGE)

`Machine Edge` sends telemetry upstream, receives OTA commands, and reports apply/rollback status.

#### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Sends telemetry over authenticated channels and retries during endpoint outage windows.

#### Inbound Interfaces

| Interface | Description |
|---|---|
| `Barista` | Daily machine operation produces brew and status telemetry. |

#### Outbound Interfaces

| Interface | Description |
|---|---|
| `Device Identity and Secrets` | Uses machine identity material for authenticated publish. |
| `Fleet Ingestion API` | Sends normalized telemetry records to ingestion API. |
| `IoT Ingest Endpoint` | Publishes telemetry payloads to cloud ingestion. |
| `MQTT Device SDK` | Uses MQTT client for edge publish and command channels. |

#### Communication Evidence

runtime: `telemetry_retry_queue` | code modules: `machine_telemetry`

### OTA Update Agent (FU-OTA-UPDATE-AGENT)

Receives campaign commands and reports update/rollback status back to `cloud control`.

#### Inbound Interfaces

none

#### Outbound Interfaces

| Interface | Description |
|---|---|
| `Device Identity and Secrets` | Uses trust anchors for firmware and endpoint authentication. |
| `Firmware Bundle Store` | Downloads signed firmware bundle. |
| `Update Campaign Orchestration` | Receives OTA rollout commands and policy constraints. |

#### Communication Evidence

runtime: `ota_campaign_schedule`, `ota_campaign_target` | code modules: `ota_update_agent`

## 6.5.2. Cloud Control (FG-CLOUD-CONTROL)

`Cloud Control` coordinates telemetry intake, campaign commands, and observability event fanout.

### Fleet Ingestion API (FU-FLEET-INGESTION-API)

Accepts machine telemetry and forwards normalized records to `observability reporting`.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| `Machine Telemetry Collection` | Sends normalized telemetry records to ingestion API. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| `Cloud Runtime SDK` | Uses cloud SDK clients for ingest and routing operations. |
| `Device Identity and Secrets` | Validates device identity and token claims. |
| `Fleet Observability Reporting` | Forwards validated telemetry and machine status events. |

#### Communication Evidence

runtime: `fleet_event_topic`, `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Consumes telemetry and OTA outcomes and emits dashboards/alerts/log events.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| `Fleet Ingestion API` | Forwards validated telemetry and machine status events. |
| `Update Campaign Orchestration` | Emits OTA rollout and rollback events for fleet visibility. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| `Cloud Logging Service` | Persists telemetry, OTA outcomes, and audit records. |
| `Device Identity and Secrets` | Uses controlled credentials for audit-log writes. |

#### Communication Evidence

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Sends OTA commands to update agents and ingests status feedback for campaign control.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| `Fleet Operator` | Defines OTA target cohorts and rollout windows. |

| Interface | Description |
| --- | --- |
| `OTA Update Agent` | Receives OTA rollout commands and policy constraints. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| `Device Identity and Secrets` | Uses trusted control-plane credentials. |
| `Firmware Signer Service` | Verifies firmware artifact signature metadata. |
| `Fleet Observability Reporting` | Emits OTA rollout and rollback events for fleet visibility. |

#### Communication Evidence

runtime: `update_campaign_orchestration` | code modules: `update_campaign_orchestrator`

### 6.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform operations coordinates runtime controls and secret distribution contracts with dependent units.

#### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Coordinates deployment/runtime controls that shape fleet service availability.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| `Platform Operator` | Operates deployment and runtime controls. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| `Cloud Logging Service` | Emits platform operation and deployment events. |
| `Device Identity and Secrets` | Requires runtime credentials and encrypted config. |

#### Communication Evidence

code modules: `cloud_runtime_ops`

#### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Provides identity validation and secure config interfaces to edge and cloud units.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| `Cloud Runtime Operations` | Requires runtime credentials and encrypted config. |

---

| Interface | Description |
| --- | --- |
| `Fleet Ingestion API` | Validates device identity and token claims. |
| `Fleet Observability Reporting` | Uses controlled credentials for audit-log writes. |
| `Machine Telemetry Collection` | Uses machine identity material for authenticated publish. |
| `OTA Update Agent` | Uses trust anchors for firmware and endpoint authentication. |
| `Security Analyst` | Audits identity trust and secret rotation controls. |
| `Update Campaign Orchestration` | Uses trusted control-plane credentials. |

### Outbound Interfaces

none

### Communication Evidence

code modules: `device_identity_secrets`

# 7. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and `platform operations`.

## 7.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

## 7.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no direct inferred runtime/deployment evidence yet.
- `Device Identity and Secrets` has no direct inferred runtime/deployment evidence yet.

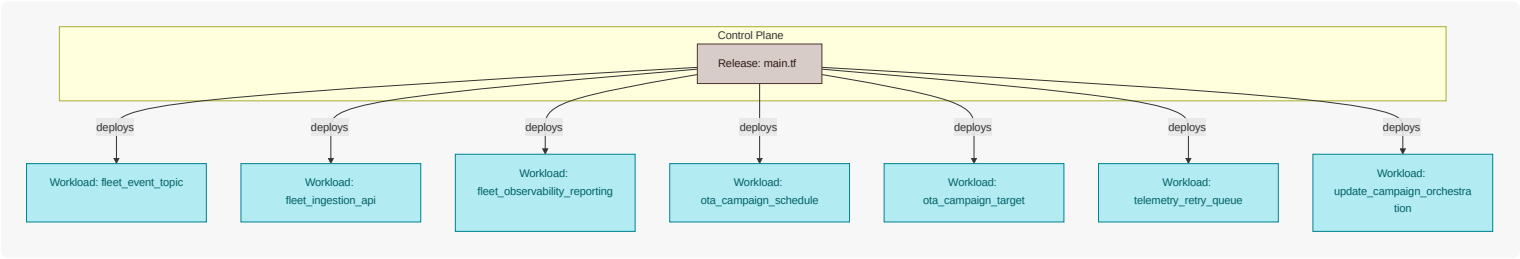
## 7.3. Recommended Next Evidence Additions

- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment control artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

## 7.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

> **NOTE**
> This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.

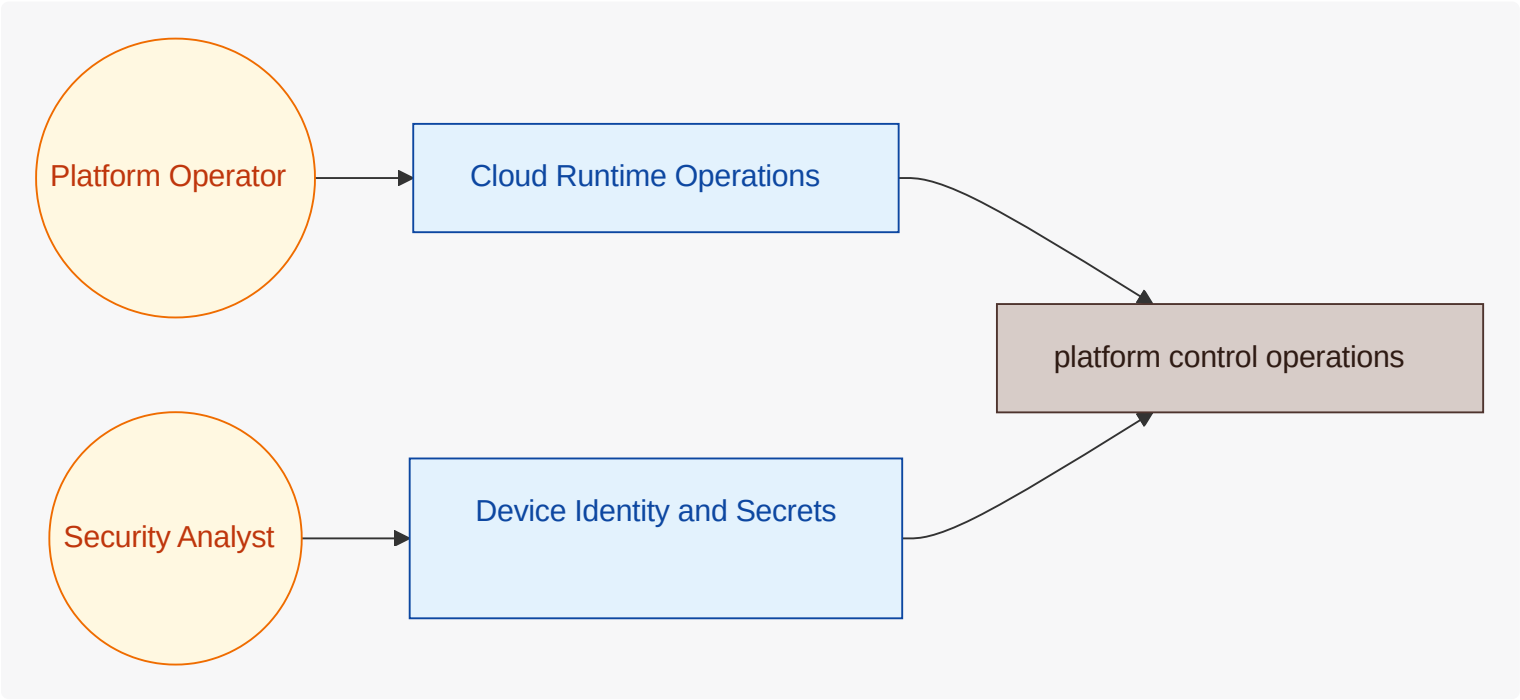



## 7.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

NOTE

Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.




## 7.6. Functional Groups and Units (View Scoped)

### 7.6.1. Machine Edge (FG-MACHINE-EDGE)

Machine-edge software artifacts are rolled independently from `cloud control` to isolate device risk.

### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

#### Machine Telemetry Collection Deployment

Deployed as edge software component with independent release cadence.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | `REQ-COF-001`, `REQ-COF-006` |
| Deployment dependencies | `Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK` |
| Inferred deployment evidence | runtime: `telemetry_retry_queue` |

### OTA Update Agent (FU-OTA-UPDATE-AGENT)

#### OTA Update Agent Deployment

Versioned and rolled independently from telemetry pipeline components.

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment trigger context | `REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005` |
| Deployment dependencies | `Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration` |
| Inferred deployment evidence | runtime: `ota_campaign_schedule`, `ota_campaign_target` |

### 7.6.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud-control units deploy independently to reduce blast radius during campaign logic changes.

#### Fleet Ingestion API (FU-FLEET-INGESTION-API)

##### Fleet Ingestion API Deployment

Deployed as independent cloud ingestion boundary to isolate scaling concerns.

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment trigger context | `REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008` |
| Deployment dependencies | `Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting` |
| Inferred deployment evidence | runtime: `fleet_event_topic`, `fleet_ingestion_api` |

#### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

##### Fleet Observability Reporting Deployment

Deployed as reporting plane with independent scaling and retention controls.

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment trigger context | `REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008` |
| Deployment dependencies | `Cloud Logging Service`, `Device Identity and Secrets` |
| Inferred deployment evidence | runtime: `fleet_observability_reporting` |

#### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

##### Update Campaign Orchestration Deployment

Isolated deployment supports safe OTA policy evolution.

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment trigger context | `REQ-COF-003`, `REQ-COF-005` |

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment dependencies | `Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting` |
| Inferred deployment evidence | runtime: `update_campaign_orchestration` |

### 7.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform operations governs release sequencing, runtime placement, and operational guardrails.

#### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

##### Cloud Runtime Operations Deployment

Owns source-to-runtime release controls and rollback guardrails.

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment trigger context | `REQ-COF-008` |
| Deployment dependencies | `Cloud Logging Service`, `Device Identity and Secrets` |
| Inferred deployment evidence | code modules: `cloud_runtime_ops` |

#### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

##### Device Identity and Secrets Deployment

Managed as shared control-plane boundary with controlled key rotation.

| Deployment Aspect | View-Scoped Detail |
| --- | --- |
| Deployment trigger context | `REQ-COF-004`, `REQ-COF-007` |
| Deployment dependencies | none |
| Inferred deployment evidence | code modules: `device_identity_secrets` |

# 8. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence.

## 8.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

## 8.2. Coverage Gaps

Coverage summary: 6/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

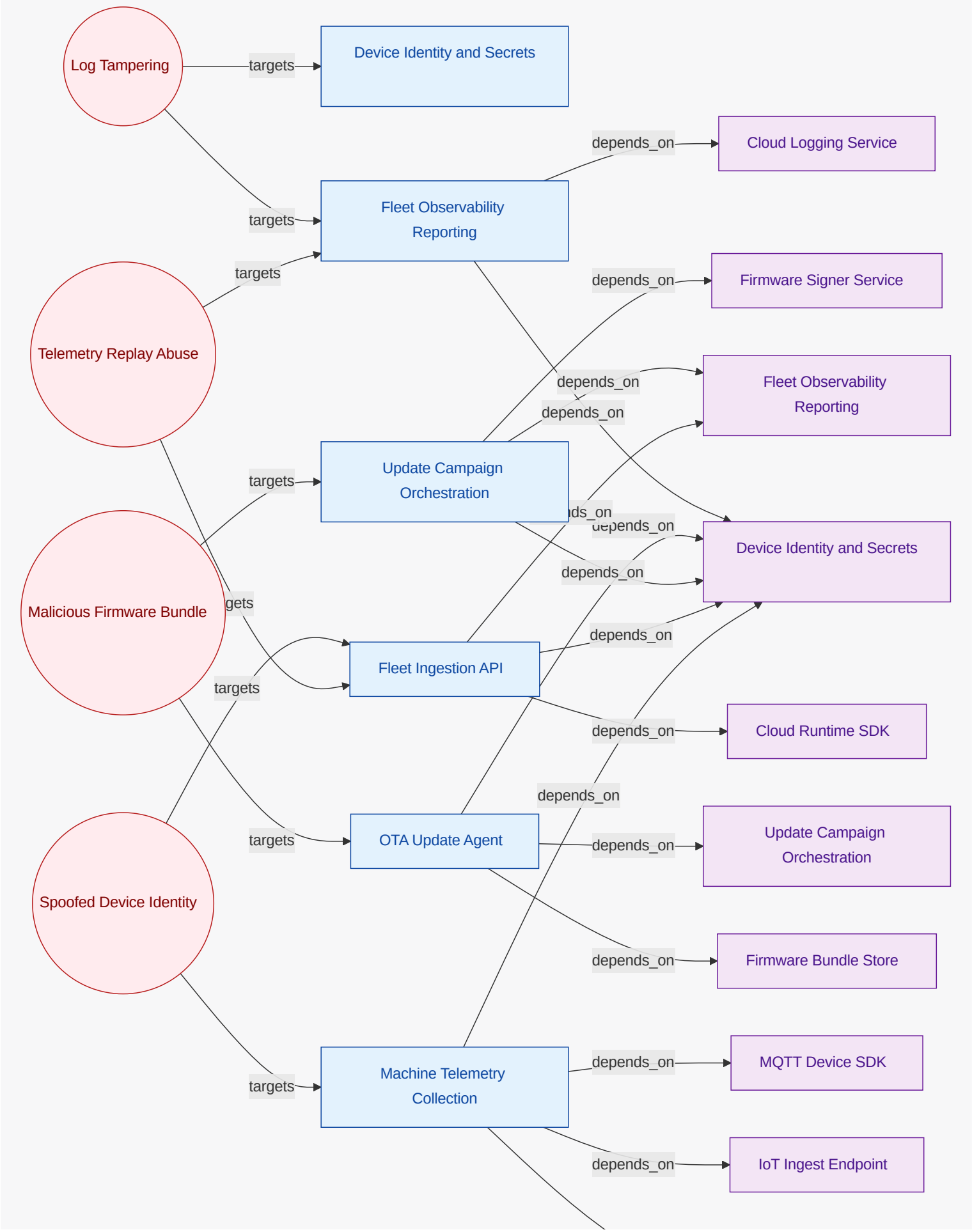
## 8.3. Recommended Next Evidence Additions

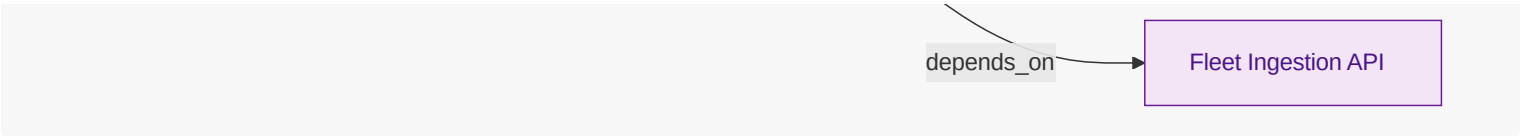
- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/events) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

## 8.4. Attack Path Diagram

What this diagram shows: authored attack vectors, their targeted units, and security-relevant dependency paths.

> **NOTE**
> Security relationships shown here are explicitly modeled or inferred from available evidence and may be incomplete. Absence of a path does not imply absence of risk. Threats that are currently mitigated (including fraud/abuse threats) should still be modeled as attack vectors for auditability and control-trace visibility.


Log Tampering
Telemetry Replay Abuse
Malicious Firmware Bundle
Spoofed Device Identity
targets
Device Identity and Secrets
Fleet Observability Reporting
Update Campaign Orchestration
Fleet Ingestion API
OTA Update Agent
Machine Telemetry Collection
depends_on
Cloud Logging Service
Firmware Signer Service
Fleet Observability Reporting
Device Identity and Secrets
Cloud Runtime SDK
Update Campaign Orchestration
Firmware Bundle Store
MQTT Device SDK
IoT Ingest Endpoint




## 8.5. Functional Groups and Units (View Scoped)

### 8.5.1. Machine Edge (FG-MACHINE-EDGE)

`Machine Edge` enforces firmware trust and authenticated telemetry at the device boundary.

#### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Protects telemetry authenticity and defends against replay abuse patterns.

##### Attack Surface

`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

##### Main Threats

`Spoofed Device Identity`

##### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

##### Security Evidence Present

runtime: `telemetry_retry_queue` | code modules: `machine_telemetry`

##### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

#### OTA Update Agent (FU-OTA-UPDATE-AGENT)

Enforces `signature valid`ation and rejects untrusted firmware bundles.

##### Attack Surface

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

##### Main Threats

`Malicious Firmware Bundle`

##### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

##### Security Evidence Present

runtime: `ota_campaign_schedule`, `ota_campaign_target` | code modules: `ota_update_agent`

##### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### 8.5.2. Cloud Control (FG-CLOUD-CONTROL)

`Cloud Control` focuses on replay detection, campaign authorization, and audit integrity.

#### Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

#### Attack Surface

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

#### Main Threats

`Spoofed Device Identity`, `Telemetry Replay Abuse`

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

runtime: `fleet_event_topic`, `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

#### Attack Surface

`Cloud Logging Service`, `Device Identity and Secrets`

#### Main Threats

`Log Tampering`, `Telemetry Replay Abuse`

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Enforces signed-firmware policy and campaign authorization boundaries.

#### Attack Surface

`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

#### Main Threats

`Malicious Firmware Bundle`

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

runtime: `update_campaign_orchestration` | code modules: `update_campaign_orchestrator`

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### 8.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform operations hardens identity, key rotation, and log integrity boundaries.

#### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Enforces operational hardening and security escalation signaling.

##### Attack Surface

`Cloud Logging Service`, `Device Identity and Secrets`

##### Main Threats

no explicit `attack vector` targeting this unit

##### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

##### Security Evidence Present

code modules: `cloud_runtime_ops`

##### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

#### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Protects credential lifecycle and enforces least-privilege access patterns.

##### Attack Surface

none

##### Main Threats

`Log Tampering`

##### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

##### Security Evidence Present

code modules: `device_identity_secrets`

##### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

# 9. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

## 9.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

## 9.2. Coverage Gaps

Coverage summary: 7/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no direct inferred runtime evidence yet.
- `Device Identity and Secrets` has no direct inferred runtime evidence yet.

## 9.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

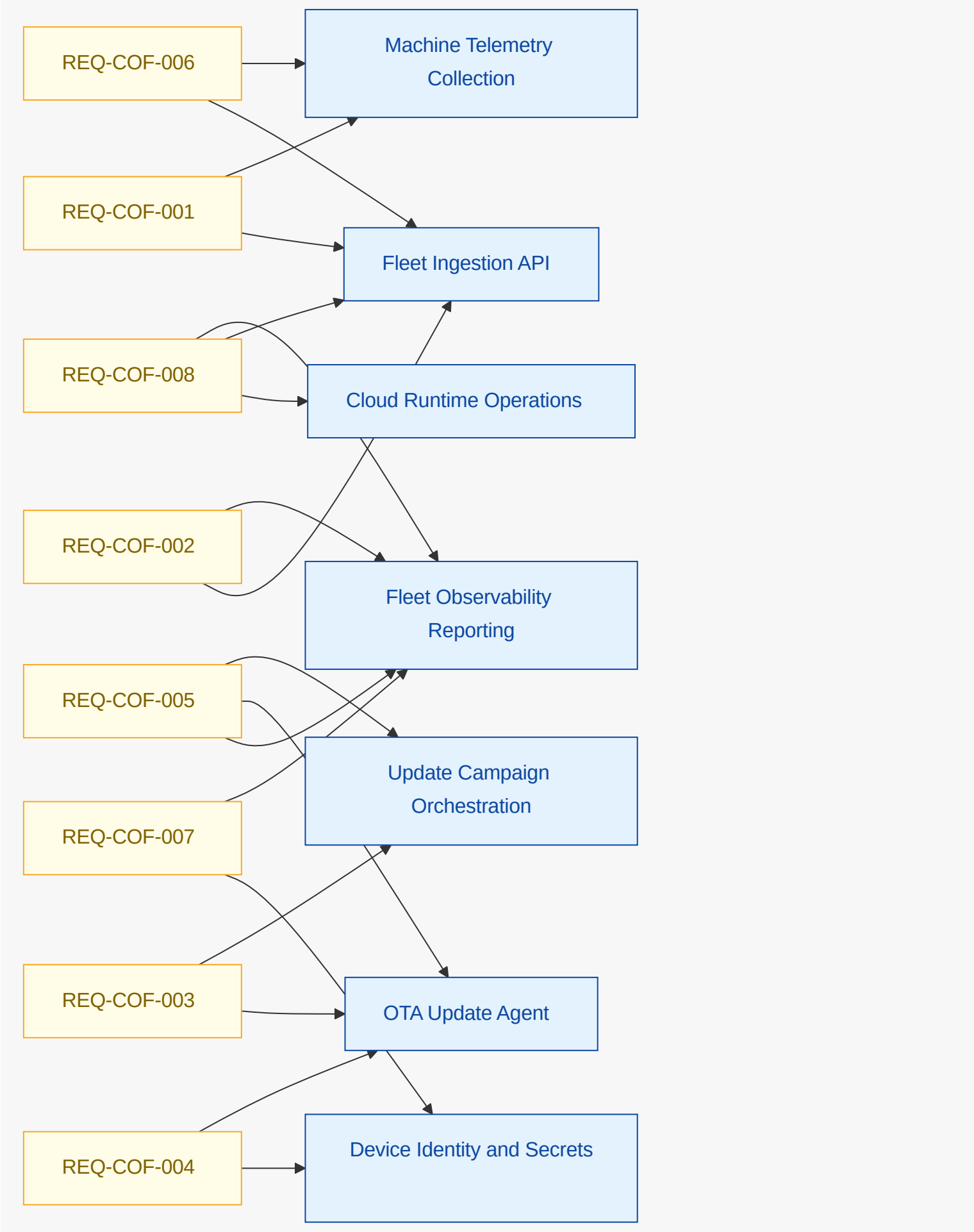
## 9.4. Traceability Evidence Graph

What this diagram shows: authored units and the inferred evidence links that support traceability confidence.

> **NOTE** This graph summarizes inferred ownership/evidence links and is intended for traceability confidence checks.

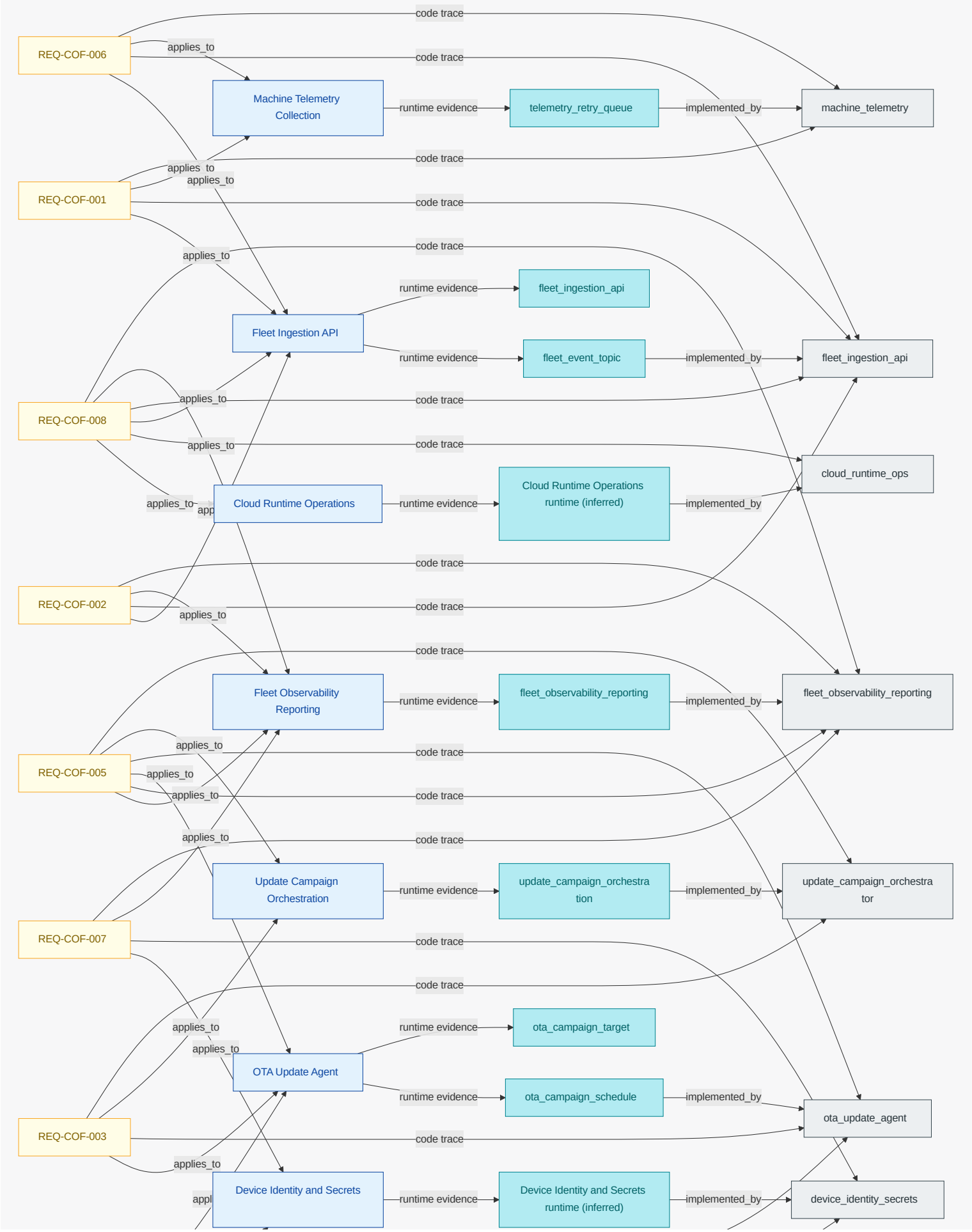
## 9.5. Requirement-to-Unit Mapping

What this diagram shows: direct authored mapping from requirements to the functional units they apply to.


REQ-COF-006
Machine Telemetry Collection
REQ-COF-001
Fleet Ingestion API
REQ-COF-008
Cloud Runtime Operations
REQ-COF-002
Fleet Observability Reporting
REQ-COF-005
Update Campaign Orchestration
REQ-COF-007
REQ-COF-003
OTA Update Agent
REQ-COF-004
Device Identity and Secrets

## 9.6. Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: requirement-to-unit links extended with inferred runtime/code evidence attached to each unit.


REQ-COF-006
REQ-COF-001
REQ-COF-008
REQ-COF-002
REQ-COF-005
REQ-COF-007
REQ-COF-003
Machine Telemetry Collection
Fleet Ingestion API
Cloud Runtime Operations
Fleet Observability Reporting
Update Campaign Orchestration
OTA Update Agent
Device Identity and Secrets
telemetry_retry_queue
fleet_ingestion_api
fleet_event_topic
Cloud Runtime Operations runtime (inferred)
fleet_observability_reporting
update_campaign_orchestration
ota_campaign_target
ota_campaign_schedule
Device Identity and Secrets runtime (inferred)
machine_telemetry
fleet_ingestion_api
cloud_runtime_ops
fleet_observability_reporting
update_campaign_orchestrator
ota_update_agent
device_identity_secrets
applies_to
code trace
runtime evidence
implemented_by

Show requirement-to-unit mappings inferred from appliesTo and authored architecture ownership boundaries.

# 10. Traceability Appendix

## 10.1. Requirement Details

### 10.1.1. REQ-COF-001

While `coffee machine fleet operation is active`, when `brew cycle completed event is received`, `the coffee fleet system` shall publish `brew telemetry` to `fleet ingestion api`. Notes: Primary machine-to-cloud telemetry contract.

### 10.1.2. REQ-COF-002

While `coffee machine fleet operation is active`, when `telemetry ingested event is received`, `the coffee fleet system` shall persist fleet telemetry metrics and logging records for `fleet operator`. Notes: Operational observability `requirement`.

### 10.1.3. REQ-COF-003

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall distribute signed firmware bundle to targeted machine cohort. Notes: OTA rollout behavior from central `cloud control`.

### 10.1.4. REQ-COF-004

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall gate firmware apply on `firmware signature is valid`. Notes: Firmware integrity and trust boundary control.

### 10.1.5. REQ-COF-005

If `firmware validation failed event is received`, then `the coffee fleet system` shall rollback firmware apply and report `ota status` to `update campaign orchestration`. Notes: Rollback and failure signaling path.

### 10.1.6. REQ-COF-006

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall queue telemetry for retry and flush when `normal mode is active`. Notes: Resilience for temporary cloud outage.

### 10.1.7. REQ-COF-007

Where `audit logging is enabled`, when `ota status reported event is received`, `the coffee fleet system` shall persist redacted `audit log record` including machine identity and campaign id. Notes: Audit/compliance logging `requirement`.

### 10.1.8. REQ-COF-008

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall throttle duplicate telemetry and notify `security analyst`. Notes: Abuse control and security response `requirement`.

## 10.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

### 10.2.1. Ota Rollout Flow

| Field | Value |
|---|---|
| ID | `VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC` |

| Field | Value |
| --- | --- |
| Kind | e2e |
| Status | partial |
| Verifies Requirements | `REQ-COF-003`, `REQ-COF-005` |
| Test Code Elements | tests/e2e/ota_rollout_flow.yaml |
| Derived Functional Owners | `FU-FLEET-OBSERVABILITY-REPORTING`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION` |
| Result Summary | not-run:1, partial:1 |
| Evidence | test-results/ota-rollback-report.md, tests/e2e/ota_rollout_flow.yaml |
| Notes | runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation |

### 10.2.2. Logging Pipeline Test

| Field | Value |
| --- | --- |
| ID | `VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF` |
| Kind | integration |
| Status | not-run |
| Verifies Requirements | `REQ-COF-002`, `REQ-COF-007` |
| Test Code Elements | tests/integration/logging_pipeline_test.ts |
| Derived Functional Owners | `FU-DEVICE-IDENTITY-SECRETS`, `FU-FLEET-INGESTION-API`, `FU-FLEET-OBSERVABILITY-REPORTING` |
| Result Summary | not-run:1, pass:1 |
| Evidence | test-results/integration-logging.json, tests/integration/logging_pipeline_test.ts |
| Notes | validates telemetry logging pipeline persistence and operational reporting signals |

### 10.2.3. Telemetry Ingest Contract Test

| Field | Value |
| --- | --- |
| ID | `VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794` |
| Kind | test |
| Status | pass |

| Field | Value |
| --- | --- |
| Verifies Requirements | `REQ-COF-001` |
| Test Code Elements | tests/contract/telemetry_ingest_contract_test.go |
| Derived Functional Owners | `FU-FLEET-INGESTION-API`, `FU-MACHINE-TELEMETRY-COLLECTION` |
| Result Summary | pass:1 |
| Evidence | test-results/contract-telemetry.json, tests/contract/telemetry_ingest_contract_test.go |
| Notes | validates telemetry ingest contract fields and accepted request semantics |

#### 10.2.4. Ota Campaign Test

| Field | Value |
| --- | --- |
| ID | `VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8` |
| Kind | unit |
| Status | pass |
| Verifies Requirements | `REQ-COF-003`, `REQ-COF-004` |
| Test Code Elements | tests/unit/ota_campaign_test.go |
| Derived Functional Owners | `FU-DEVICE-IDENTITY-SECRETS`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION` |
| Result Summary | pass:2 |
| Evidence | test-results/unit-ota.xml, tests/unit/ota_campaign_test.go |
| Notes | verifies OTA campaign `signature valid`ation and rollout trigger behavior |

## 10.3. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

| Check ID | Requirement | Result | Evidence / Notes |
| --- | --- | --- | --- |
| `VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794` | `REQ-COF-001` | pass | test-results/contract-telemetry.json<br>ingest payload accepted with machine identity and telemetry fields |

| Check ID | Requirement | Result | Evidence / Notes |
|---|---|---|---|
| `VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF` | `REQ-COF-002` | not-run | none<br>No parsed test-result artifact for this `requirement` yet. |
| `VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC` | `REQ-COF-003` | not-run | none<br>No parsed test-result artifact for this `requirement` yet. |
| `VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8` | `REQ-COF-003` | pass | test-results/unit-ota.xml |
| `VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8` | `REQ-COF-004` | pass | test-results/unit-ota.xml |
| `VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC` | `REQ-COF-005` | partial | test-results/ota-rollback-report.md |
| `VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF` | `REQ-COF-007` | pass | test-results/integration-logging.json<br>`audit log record` persisted with redacted machine identity |

# 11. Reference Index

## 11.1. Authored References

### ACT-BARISTA

Kind: Actor Name: Barista

Operates coffee machines and observes local device status.

### ACT-FLEET-OPERATOR

Kind: Actor Name: Fleet Operator

Manages OTA campaigns and fleet-wide operational decisions.

### ACT-PLATFORM-OPERATOR

Kind: Actor Name: Platform Operator

Operates cloud runtime, deployment, and reliability controls.

### ACT-SECURITY-ANALYST

Kind: Actor Name: Security Analyst

Reviews firmware integrity, identity controls, and audit logging.

### AV-LOG-TAMPERING

Kind: Attack Vector Name: Log Tampering

Attempts to suppress or alter audit/operational logs.

### AV-MALICIOUS-FIRMWARE-BUNDLE

Kind: Attack Vector Name: Malicious Firmware Bundle

Tampered OTA payload targeting update agent execution.

### AV-SPOOFED-DEVICE-IDENTITY

Kind: Attack Vector Name: Spoofed Device Identity

Unauthorized identity attempting telemetry/command access.

### AV-TELEMETRY-REPLAY-ABUSE

Kind: Attack Vector Name: Telemetry Replay Abuse

Replayed telemetry intended to overload ingestion and corrupt insights.

### FG-CLOUD-CONTROL

Kind: Functional Group Name: Cloud Control

Central cloud ingestion, campaign control, and fleet reporting.

### FG-MACHINE-EDGE

Kind: Functional Group Name: Machine Edge

Device-local telemetry and OTA execution responsibilities.

### FG-PLATFORM-OPERATIONS

Kind: Functional Group Name: Platform Operations

Cloud runtime operations plus identity and secret-management controls.

### FU-CLOUD-RUNTIME-OPERATIONS

Kind: Functional Unit Name: Cloud Runtime Operations

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

### FU-DEVICE-IDENTITY-SECRETS

Kind: Functional Unit Name: Device Identity and Secrets

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

### FU-FLEET-INGESTION-API

Kind: Functional Unit Name: Fleet Ingestion API

Receives telemetry and machine status, applies validation, and emits ingestion events.

### FU-FLEET-OBSERVABILITY-REPORTING

Kind: Functional Unit Name: Fleet Observability Reporting

Aggregates telemetry and OTA outcomes into dashboards, alerts, and audit records.

### FU-MACHINE-TELEMETRY-COLLECTION

Kind: Functional Unit Name: Machine Telemetry Collection

Collects brew/device telemetry and transmits cloud-bound telemetry payloads.

### FU-OTA-UPDATE-AGENT

Kind: Functional Unit Name: OTA Update Agent

Downloads signed firmware bundles, validates integrity, applies updates, and reports status.

### FU-UPDATE-CAMPAIGN-ORCHESTRATION

Kind: Functional Unit Name: Update Campaign Orchestration

Plans OTA rollout cohorts, drives update commands, and handles rollback flow decisions.

### REF-CLOUD-LOGGING-SERVICE

Kind: Referenced Element Name: Cloud Logging Service

platform_service / runtime

### REF-CLOUD-RUNTIME-SDK

Kind: Referenced Element Name: Cloud Runtime SDK

third_party_library / code

### REF-FIRMWARE-BUNDLE-STORE

Kind: Referenced Element Name: Firmware Bundle Store

object_store / runtime

### REF-FIRMWARE-SIGNER-SERVICE

Kind: Referenced Element Name: Firmware Signer Service

signing_service / runtime

### REF-IOT-INGEST-ENDPOINT

Kind: Referenced Element Name: IoT Ingest Endpoint

external_service_endpoint / runtime

### REF-MQTT-DEVICE-SDK

Kind: Referenced Element Name: MQTT Device SDK

third_party_library / code

### REQ-COF-001

Kind: Requirement Name: REQ-COF-001 Resolves to: `canonical` While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api.

### REQ-COF-002

Kind: Requirement Name: REQ-COF-002 Resolves to: `canonical` While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics and logging records for fleet operator.

### REQ-COF-003

Kind: Requirement Name: REQ-COF-003 Resolves to: `canonical` Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort.

### REQ-COF-004

Kind: Requirement Name: REQ-COF-004 Resolves to: `canonical` Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall gate firmware apply on firmware signature is valid.

### REQ-COF-005

Kind: Requirement Name: REQ-COF-005 Resolves to: `canonical` If firmware validation failed event is received, then the coffee fleet system shall rollback firmware apply and report ota status to update campaign orchestration.

### REQ-COF-006

Kind: Requirement Name: REQ-COF-006 Resolves to: `canonical` If cloud ingest endpoint is unavailable, then the coffee fleet system shall queue telemetry for retry and flush when normal mode is active.

### REQ-COF-007

Kind: Requirement Name: REQ-COF-007 Resolves to: `canonical` Where audit logging is enabled, when ota status reported event is received, the coffee fleet system shall persist redacted audit log record including machine identity and campaign id.

### REQ-COF-008

Kind: Requirement Name: REQ-COF-008 Resolves to: `canonical` If telemetry replay abuse is detected, then the coffee fleet system shall throttle duplicate telemetry and notify security analyst.

## 11.2. Catalog References

### barista

Operates coffee machines and observes local machine health. Canonical: `open`

### fleet operator

Owns OTA rollout campaigns and fleet-level reliability. Canonical: `open`

### platform operator

Owns runtime deployment and operational controls. Canonical: `open`

### security analyst

Reviews identity, firmware integrity, and logging controls. Canonical: `open`

### audit logging is enabled

Feature flag enabling audit-grade log persistence. Canonical: `open`

### log tampering

Attempts to suppress or alter security/operational log records. Canonical: `open`

### brew cycle completed event is received

Event raised when a coffee machine completes a brew cycle. Canonical: `open`

### update campaign orchestration

Selects target machine groups and coordinates OTA rollout phases. Canonical: `open`

### ota campaign scheduled event is received

Event raised when a fleet OTA campaign is scheduled. Canonical: `open`

### cloud control

Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Canonical: `open`

### cloud runtime sdk

SDK used by cloud units for runtime integrations. Canonical: `open`

### machine edge

Capability area for device-local telemetry and OTA execution behavior. Canonical: `open`

### device identity and secrets

Manages device identity material and secure configuration. Canonical: `open`

### spoofed device identity

Unauthorized actor impersonates a machine identity. Canonical: `open`

### ota update agent

Downloads, validates, and applies signed firmware updates on machines. Canonical: `open`

### firmware bundle store

Object store containing signed OTA firmware bundles. Canonical: `open`

### malicious firmware bundle

Tampered firmware payload attempts to execute on devices. Canonical: `open`

### fleet operator

Owns OTA rollout campaigns and fleet-level reliability. Canonical: `open`

### coffee machine fleet operation is active

State where machines and central cloud are actively serving operations. Canonical: `open`

### platform operations

Capability area for runtime operations and device identity/secrets lifecycle. Canonical: `open`

### cloud ingest endpoint is unavailable

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion. Canonical: `open`

### fleet ingestion api

Ingests telemetry and machine status from edge devices. Canonical: `open`

### iot ingest endpoint

Cloud endpoint for receiving telemetry and status from devices. Canonical: `open`

### cloud logging service

Managed log and metrics service used for fleet observability. Canonical: `open`

### mqtt device sdk

SDK used by edge components for telemetry and command transport. Canonical: `open`

### normal mode is active

Default operation mode for telemetry and OTA behavior. Canonical: `open`

### fleet observability reporting

Aggregates telemetry and update outcomes into operational and audit signals. Canonical: `open`

### ota campaign is enabled

Feature flag enabling centralized OTA rollout behavior. Canonical: `open`

### ota status reported event is received

Event raised when device reports OTA apply status. Canonical: `open`

### platform operator

Owns runtime deployment and operational controls. Canonical: `open`

### cloud runtime operations

Operates cloud runtime deployment and scaling controls. Canonical: `open`

### security analyst

Reviews identity, firmware integrity, and logging controls. Canonical: `open`

### firmware signature is valid

Condition indicating firmware signature verification succeeded. Canonical: `open`

### firmware signer service

Service that signs firmware artifacts before OTA rollout. Canonical: `open`

### barista

Operates coffee machines and observes local machine health. Canonical: `open`

### machine telemetry collection

Collects brew/device telemetry and publishes cloud-bound payloads. Canonical: `open`

### telemetry ingested event is received

Event raised when cloud ingestion accepts telemetry. Canonical: `open`

### telemetry replay abuse

Replayed telemetry floods ingestion and distorts reporting. Canonical: `open`

### telemetry replay abuse is detected

Event raised when replayed telemetry behavior is detected and flagged as abuse. Canonical: `open`

### coffee fleet system

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture. Canonical: `open`

### firmware validation failed event is received

Event raised when signature/hash validation fails on firmware. Canonical: `open`

### log tampering

Attempts to suppress or alter security/operational log records. Canonical: `open`

### malicious firmware bundle

Tampered firmware payload attempts to execute on devices. Canonical: `open`

### spoofed device identity

Unauthorized actor impersonates a machine identity. Canonical: `open`

### telemetry replay abuse

Replayed telemetry floods ingestion and distorts reporting. Canonical: `open`

### firmware signature is valid

Condition indicating firmware signature verification succeeded. Canonical: `open`

### audit log record

Immutable log record for security and compliance analysis. Canonical: `open`

### brew telemetry

Machine telemetry including brew duration, temperature, and status metadata. Canonical: `open`

### ota campaign policy

Policy describing rollout cohorts, windows, and rollback criteria. Canonical: `open`

### ota status

Device-level status for firmware download, apply, and rollback outcomes. Canonical: `open`

### brew cycle completed event is received

Event raised when a coffee machine completes a brew cycle. Canonical: `open`

### cloud ingest endpoint is unavailable

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion. Canonical: `open`

### firmware validation failed event is received

Event raised when signature/hash validation fails on firmware. Canonical: `open`

### ota campaign scheduled event is received

Event raised when a fleet OTA campaign is scheduled. Canonical: `open`

### ota status reported event is received

Event raised when device reports OTA apply status. Canonical: `open`

### telemetry ingested event is received

Event raised when cloud ingestion accepts telemetry. Canonical: `open`

### telemetry replay abuse is detected

Event raised when replayed telemetry behavior is detected and flagged as abuse. Canonical: `open`

### audit logging is enabled

Feature flag enabling audit-grade log persistence. Canonical: `open`

### ota campaign is enabled

Feature flag enabling centralized OTA rollout behavior. Canonical: `open`

### cloud control

Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Canonical: `open`

### machine edge

Capability area for device-local telemetry and OTA execution behavior. Canonical: `open`

### platform operations

Capability area for runtime operations and device identity/secrets lifecycle. Canonical: `open`

### cloud runtime operations

Operates cloud runtime deployment and scaling controls. Canonical: `open`

### device identity and secrets

Manages device identity material and secure configuration. Canonical: `open`

### fleet ingestion api

Ingests telemetry and machine status from edge devices. Canonical: `open`

### fleet observability reporting

Aggregates telemetry and update outcomes into operational and audit signals. Canonical: `open`

### machine telemetry collection

Collects brew/device telemetry and publishes cloud-bound payloads. Canonical: `open`

### ota update agent

Downloads, validates, and applies signed firmware updates on machines. Canonical: `open`

### update campaign orchestration

Selects target machine groups and coordinates OTA rollout phases. Canonical: `open`

### normal mode is active

Default operation mode for telemetry and OTA behavior. Canonical: `open`

### cloud logging service

Managed log and metrics service used for fleet observability. Canonical: `open`

### cloud runtime sdk

SDK used by cloud units for runtime integrations. Canonical: `open`

### firmware bundle store

Object store containing signed OTA firmware bundles. Canonical: `open`

### firmware signer service

Service that signs firmware artifacts before OTA rollout. Canonical: `open`

### iot ingest endpoint

Cloud endpoint for receiving telemetry and status from devices. Canonical: `open`

### mqtt device sdk

SDK used by edge components for telemetry and command transport. Canonical: `open`

### coffee machine fleet operation is active

State where machines and central cloud are actively serving operations. Canonical: `open`

### coffee fleet system

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture. Canonical: `open`

### actor

A person or operational role that interacts with functional units. Canonical: `open`

### attack vector

A technical misuse or attack path that targets functional, referenced, or runtime elements. Canonical: `open`

### authored mapping

An explicit relationship declared in architecture inputs between authored or referenced elements. Canonical: `open`

### code element

An inferred code structure or ownership element discovered from source trees and build metadata. Canonical: `open`

### functional group

A major authored capability area that groups related functional units. Canonical: `open`

### functional unit

An authored working unit inside a functional group that owns specific behavior. Canonical: `open`

### inferred mapping

A discovered relationship that links inferred runtime/code elements upward to authored design. Canonical: `open`

### referenced element

An architecture-relevant external, platform, or third-party dependency represented by role. Canonical: `open`

### requirement

A structured requirement statement that defines expected system behavior and maps to functional ownership. Canonical: `open`

### runtime element

An inferred runtime realization element discovered from infrastructure and deployment sources. Canonical: `open`

### upward linking

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. Canonical: `open`

### verification check

An inferred or authored verification artifact that validates requirement behavior with test evidence. Canonical: `open`

## 11.3. Inferred Runtime References

| ID | Kind | Owner | Description | Source |
|---|---|---|---|---|
| ota_campaign_schedule | Runtime eventbridge_rule | FU-OTA-UPDATE-AGENT | schedules OTA campaign orchestration triggers for configured rollout windows | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |
| ota_campaign_target | Runtime eventbridge_target | FU-OTA-UPDATE-AGENT | routes scheduled OTA campaign events to the orchestration runtime handler | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |
| fleet_ingestion_api | Runtime lambda_function | FU-FLEET-INGESTION-API | receives device telemetry, validates payload shape, and publishes normalized fleet events | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |
| fleet_observability_reporting | Runtime lambda_function | FU-FLEET-OBSERVABILITY-REPORTING | aggregates telemetry outcomes into fleet operations and security reporting signals | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |

| ID | Kind | Owner | Description | Source |
|---|---|---|---|---|
| update_campaign_orchestration | Runtime lambda_function | FU-UPDATE-CAMPAIGN-ORCHESTRATION | coordinates OTA campaign rollout windows and device cohort targeting | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |
| telemetry_retry_queue | Runtime queue | FU-MACHINE-TELEMETRY-COLLECTION | buffers telemetry during ingest outages for retry once normal mode resumes | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |
| fleet_event_topic | Runtime topic | FU-FLEET-INGESTION-API | broadcasts fleet-wide telemetry and OTA lifecycle events to subscribed processors | examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf |

## 11.4. Inferred Code References

| ID | Kind | Owner | Description | Source |
|---|---|---|---|---|
| cloud_runtime_ops.go | Code source_file | FU-CLOUD-RUNTIME-OPERATIONS | wires cloud runtime handlers and integration entrypoints | cloud_runtime_ops.go |
| device_identity_secrets.ts | Code source_file | FU-DEVICE-IDENTITY-SECRETS | resolves device identity credentials and firmware verification secrets | device_identity_secrets.ts |

| ID | Kind | Owner | Description | Source |
|---|---|---|---|---|
| fleet_ingestion_api.go | Code source_file | FU-FLEET-INGESTION-API | ingests device telemetry and emits normalized fleet ingestion events | fleet_ingestion_api.go |
| fleet_observability_reporting.ts | Code source_file | FU-FLEET-OBSERVABILITY-REPORTING | aggregates telemetry and OTA outcome signals for fleet observability reporting | fleet_observability_reporting.ts |
| machine_telemetry.rs | Code source_file | FU-MACHINE-TELEMETRY-COLLECTION | builds machine telemetry and status payloads at the edge | machine_telemetry.rs |
| ota_update_agent.rs | Code source_file | FU-OTA-UPDATE-AGENT | applies OTA campaign updates and reports rollout outcomes | ota_update_agent.rs |
| update_campaign_orchestrator.ts | Code source_file | FU-UPDATE-CAMPAIGN-ORCHESTRATION | plans OTA rollout cohorts and campaign lifecycle transitions | update_campaign_orchestrator.ts |
| CODE-HANDLEROLLBACK | Code symbol | FU-UPDATE-CAMPAIGN-ORCHESTRATION | n/a | update_campaign_orchestrator.ts:10 |
| CODE-INGESTTELEMETRY | Code symbol | FU-FLEET-INGESTION-API | n/a | fleet_ingestion_api.go:11 |
| CODE-NOTIFYREPLAYABUSE | Code symbol | FU-FLEET-OBSERVABILITY-REPORTING | n/a | fleet_observability_reporting.ts:15 |

| ID | Kind | Owner | Description | Source |
|---|---|---|---|---|
| CODE-PERSISTAUDITLOG | Code symbol | FU-FLEET-OBSERVABILITY-REPORTING | n/a | fleet_observability_reporting.ts:10 |
| CODE-PERSISTTELEMETRYMETRIC | Code symbol | FU-FLEET-OBSERVABILITY-REPORTING | n/a | fleet_observability_reporting.ts:5 |
| CODE-PLANCAMPAIGN | Code symbol | FU-UPDATE-CAMPAIGN-ORCHESTRATION | n/a | update_campaign_orchestrator.ts:5 |
| CODE-PUBLISHBREWTELEMETRY | Code symbol | FU-MACHINE-TELEMETRY-COLLECTION | n/a | machine_telemetry.rs:10 |
| CODE-QUEUEFORRETRY | Code symbol | FU-MACHINE-TELEMETRY-COLLECTION | n/a | machine_telemetry.rs:15 |
| CODE-RAISESECURITYNOTIFICATION | Code symbol | FU-CLOUD-RUNTIME-OPERATIONS | n/a | cloud_runtime_ops.go:6 |
| CODE-REDACTIDENTITYFORAUDIT | Code symbol | FU-DEVICE-IDENTITY-SECRETS | n/a | device_identity_secrets.ts:10 |
| CODE-REJECTREPLAY | Code symbol | FU-FLEET-INGESTION-API | n/a | fleet_ingestion_api.go:16 |
| CODE-ROLLBACKANDREPORT | Code symbol | FU-OTA-UPDATE-AGENT | n/a | ota_update_agent.rs:14 |
| CODE-VALIDATEANDAPPLY | Code symbol | FU-OTA-UPDATE-AGENT | n/a | ota_update_agent.rs:5 |
| CODE-VERIFYFIRMWARESIGNATURE | Code symbol | FU-DEVICE-IDENTITY-SECRETS | n/a | device_identity_secrets.ts:5 |

## 11.5. Verification References

| ID | Kind | Status | Description | Source | Canonical |
|---|---|---|---|---|---|
| VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC | e2e | partial | runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation | test-results/ota-rollback-report.md (+1) | canonical |
| VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF | integration | not-run | validates telemetry logging pipeline persistence and operational reporting signals | test-results/integration-logging.json (+1) | canonical |
| VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794 | test | pass | validates telemetry ingest contract fields and accepted request semantics | test-results/contract-telemetry.json (+1) | canonical |
| VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8 | unit | pass | verifies OTA campaign signature validation and rollout trigger behavior | test-results/unit-ota.xml (+1) | canonical |